



Lesson 6: k-NN, Naive Bayes and Support Vector Machines

Technologies for Artificial Intelligence

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Before we start

- Exercise 5, which we discuss now
- The score that was too good, and what the honest number cost you

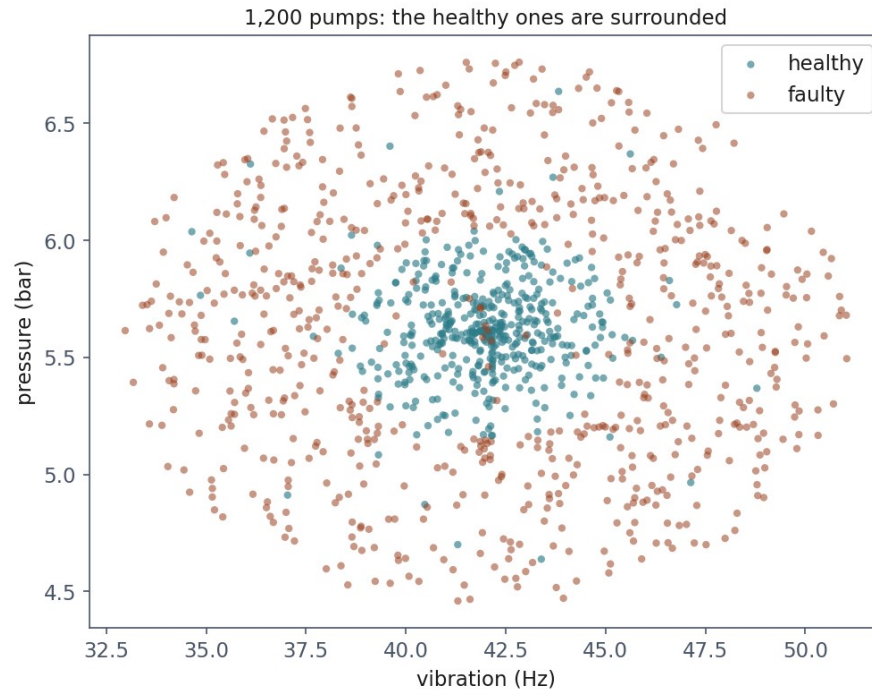
Today: the first lesson that offers a choice

- **k-nearest neighbours (k-NN)**: remember, and vote
- The **curse of dimensionality**: the bill for that
- **Naive Bayes**: one strong assumption, bought cheap
- **Support vector machines (SVM)**: margins, and kernels
- Which family to reach for, and why that beats tuning

1,200 pumps, two readings each

- Vibration in Hz, pressure in bar
- A pump is faulty when its readings fall **outside the design envelope**, too low as readily as too high
- **61.3%** faulty; 4% of the labels are deliberately flipped

The shape of the problem



What a straight boundary costs

Model	Cross-validated accuracy
Always predict the majority class	0.613
Logistic regression (lesson 4)	0.613 $\pm$ 0.000

k-nearest neighbours: the whole algorithm

- To classify a new point, find the **k closest** training points and take the majority vote
- There is no training step: “fitting” means **storing the data**
- Called a **lazy learner**: an unusually honest name for an algorithm

Distance is all it has

$$d(x, x') = \sqrt{\sum_{j=1}^n (x_j - x'_j)^2}$$

Scale first, or vibration decides everything

- Vibration runs over **tens of Hz**; pressure over **a couple of bar**
- Unscaled, a one-bar pressure difference is invisible beside a ten-Hz one
- Lesson 2's scaling argument, arriving with **immediate consequences**

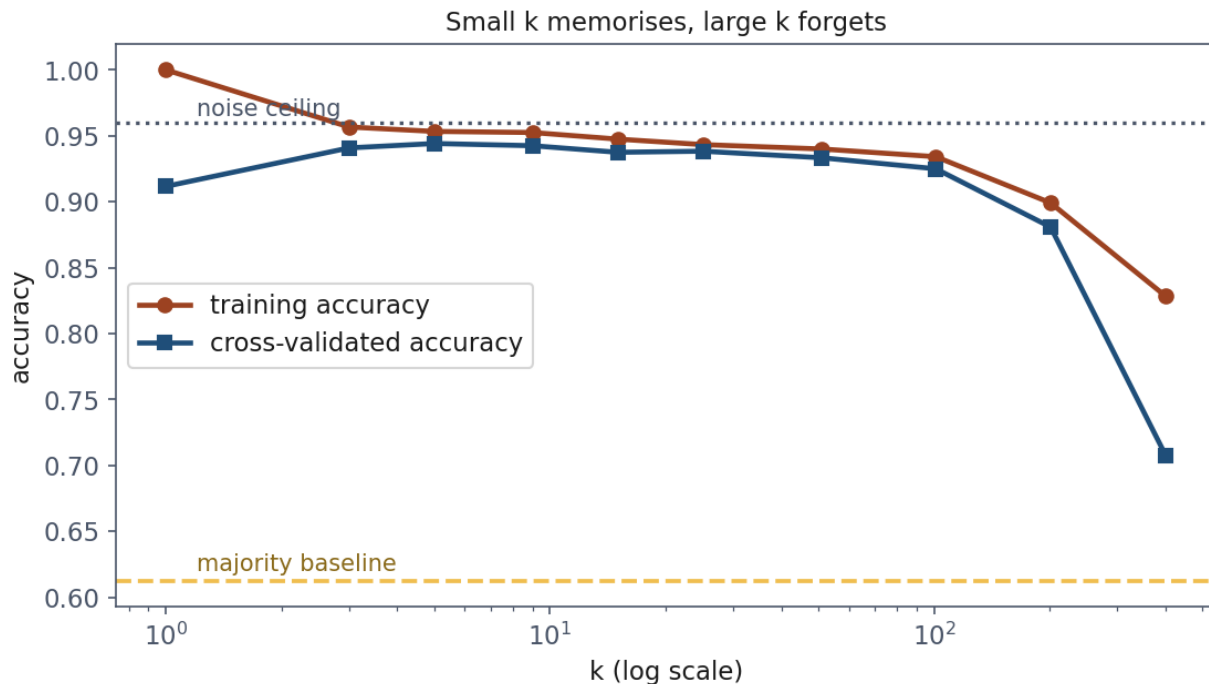
k is the bias-variance dial, made visible

- **Small k**: the boundary follows every point, mislabelled ones included. Low bias, high variance
- **Large k**: the vote is taken over a wide neighbourhood, and eventually stops following real structure. High bias, low variance
- At $k = 1$ the training accuracy is **exactly 1.000**. Always, on any dataset

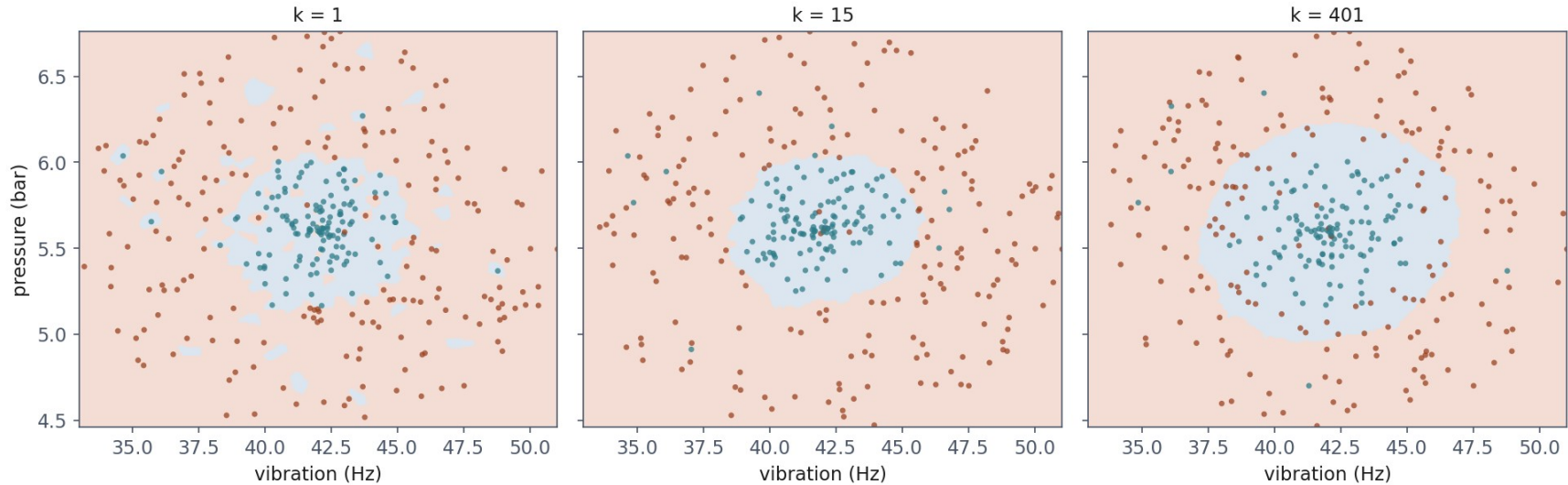
Measured on the pumps

k	Training	Cross-validated
1	1.000	0.912
5	0.953	0.944
15	0.948	0.938
51	0.940	0.933
401	0.828	0.708

The two curves, and the two lines that bound them



The same data at three values of k



What k-NN costs

- No training time at all: you pay at **every prediction** instead
- A naive implementation compares the query against **every** training row: $O(mn)$ for m rows and n features
- **The model is the dataset.** You cannot ship one without the other

Now the price: add columns of nothing

- Extra columns of **pure noise**, drawn from a normal distribution
- The two real readings are **untouched**: the problem is exactly as solvable
- Only the number of columns changes

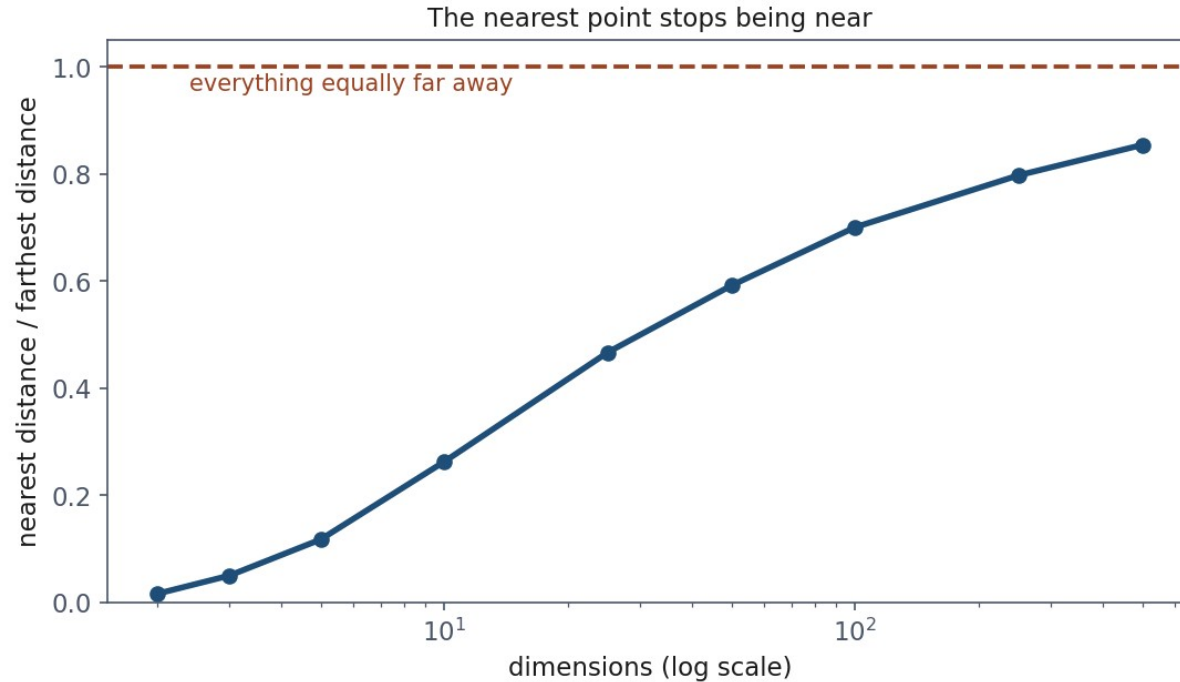
The signal never left

Total columns	Accuracy	Above baseline
2	0.938	+0.325
12	0.762	+0.149
27	0.662	+0.049
52	0.602	-0.011
102	0.578	-0.035

Why: distances stop varying

- Scatter points in a cube, pick one, measure its **nearest** and its **farthest** neighbour
- In two dimensions those are very different numbers, which is what makes “nearest” a meaningful word
- Each new dimension adds its own squared difference, and the contributions **average out**

The nearest point stops being near



Nearest divided by farthest

Dimensions	nearest ÷ farthest
2	0.016
10	0.263
50	0.592
100	0.701
500	0.855

In 100 dimensions the nearest point is 70% as far as the farthest

- In **two** dimensions it is **2%**: “nearest” picks out something special
- In **one hundred** it is **70%**: the nearest point is barely nearer than a random one
- A vote among “the five nearest” becomes a vote among five taken at random

What the curse is, and is not

- **Not** that high-dimensional problems are unlearnable: lesson 9's networks work in thousands of dimensions
- That **methods built on distance lose their footing**, because the quantity they depend on stops varying
- k-NN, k-means (lesson 8), and radial basis function (RBF) kernels, all of them

Notebook 1, live

- k-NN on the pumps, choosing k, and the curse measured

Break

- Twelve minutes

Naive Bayes: turn the question around

- We want the probability of the class **given** the readings
- The prior is a count; the denominator is identical across classes and cancels
- Everything hard sits in one term: the probability of **this exact combination of readings** among examples of that class

Bayes' rule, applied to a class

$$P(y = c \mid x) = \frac{P(x \mid y = c) P(y = c)}{P(x)}$$

The decision drops the denominator

$$\hat{y} = \arg \max_c P(x | y = c) P(y = c)$$

The assumption: independent, given the class

- **Given the class, the features are independent of one another**
- The joint density then factorises into n one-dimensional densities
- Each factor is estimated from the rows of that class alone

The factorisation the assumption
buys

$$P(x \mid y = c) = \prod_{j=1}^n P(x_j \mid y = c)$$

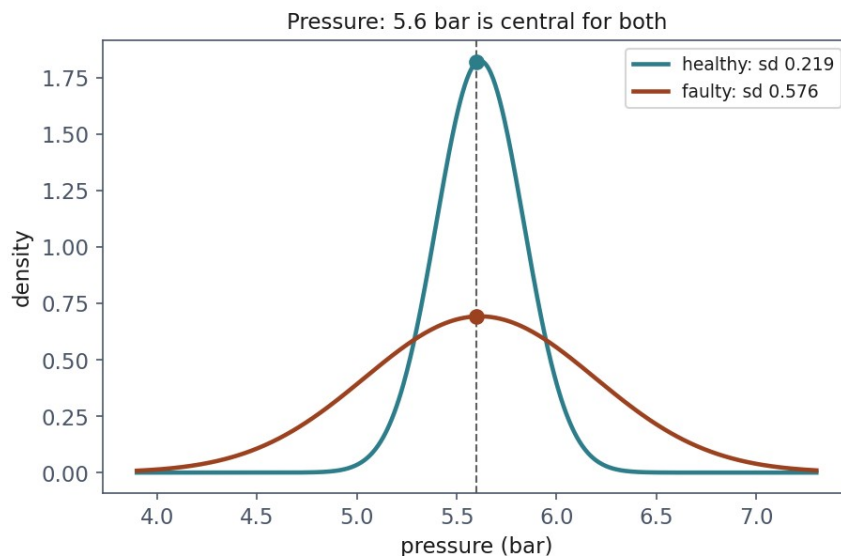
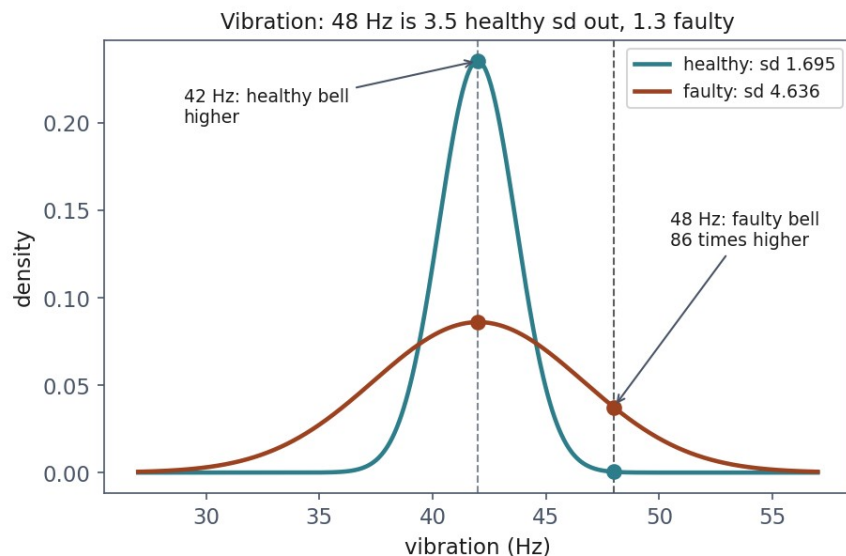
Gaussian: each factor is a bell curve

$$P(x_j \mid y = c) = \frac{1}{\sqrt{2\pi\sigma_{jc}^2}} \exp \left[-\frac{(x_j - \mu_{jc})^2}{2\sigma_{jc}^2} \right]$$

Ten numbers, and the widths decide

Class	Prior	Vibration mean	Vibration sd	Pressure mean	Pressure sd
Healthy	0.3875	41.986 Hz	1.695	5.617 bar	0.219
Faulty	0.6125	42.015 Hz	4.636	5.616 bar	0.576

48 Hz: one bell is 86 times the other



Add the logs, do not multiply

$$\hat{y} = \arg \max_c \left[\log P(y = c) + \sum_j \log P(x_j \mid y = c) \right]$$

48 Hz, 5.6 bar: faulty, 0.981

Class	log prior	log P(vibration)	log P(pressure)	Total
Healthy	-0.948	-7.745	+0.598	-8.095
Faulty	-0.490	-3.286	-0.368	-4.144

From the two totals to 0.981

- Each total is a log score: $\log \text{prior} + \log P(\text{vibration}) + \log P(\text{pressure})$
- The gap: $-4.144 - (-8.095) = \mathbf{3.951}$, in favour of faulty
- $P(\text{faulty}) = 1 / (1 + e^{(-3.951)}) = 1 / 1.0192 = \mathbf{0.981}$
- As odds: $e^{3.951} \approx 52$, so **52 to 1**, and $52 / 53 = 0.981$

A wonderful bargain, and almost never true

- Training is **a single pass**; adding features costs almost nothing
- But vibration and pressure are both driven by the operating point
- “New” is not independent of “York” given the topic. Symptoms co-occur
- So the question is not whether it holds: it is **when being wrong about it costs you nothing**

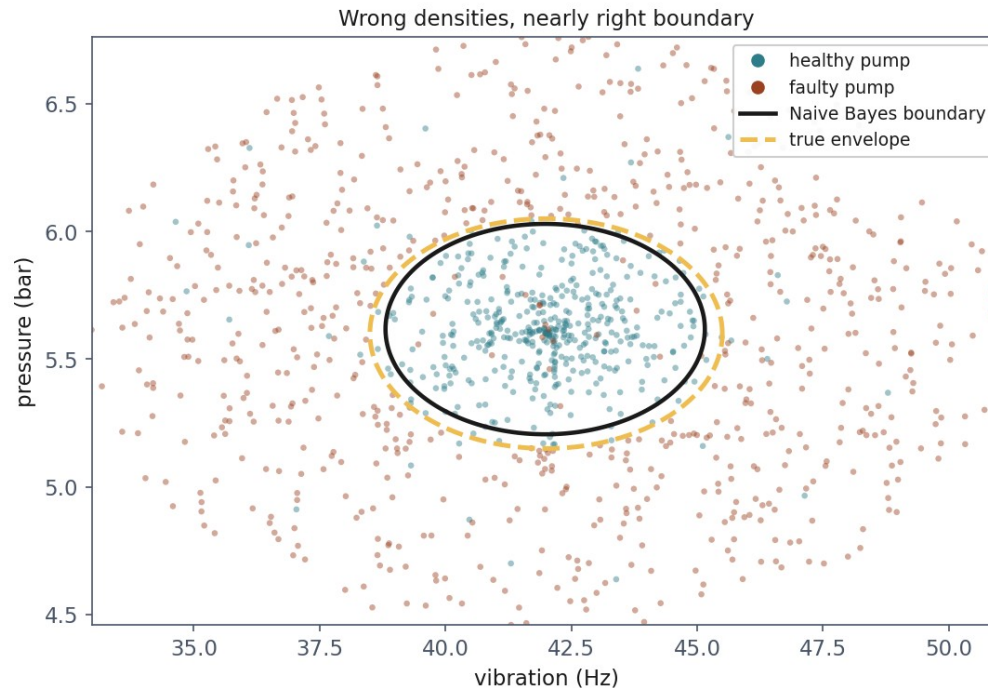
On the pumps, it scores 0.933

Model	Accuracy
Majority baseline	0.613
Logistic regression	0.613
Gaussian Naive Bayes	0.933
k-NN, $k = 5$	0.944

Uncorrelated, but not independent

Within class	Readings	Squared distances
healthy pumps	-0.006	+0.168
faulty pumps	-0.049	-0.420

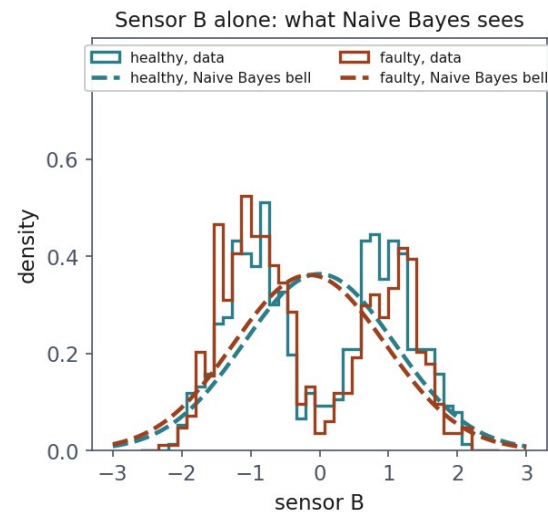
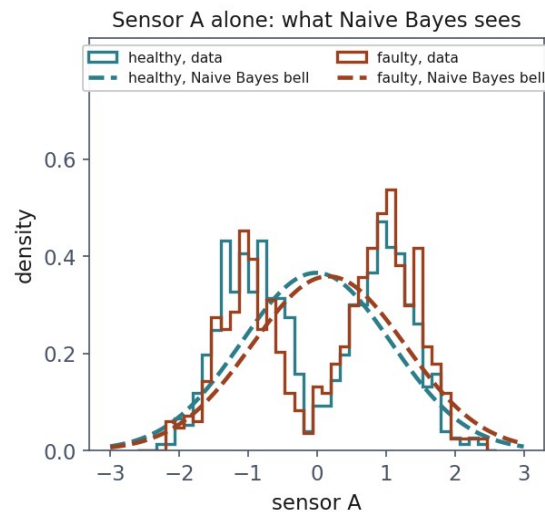
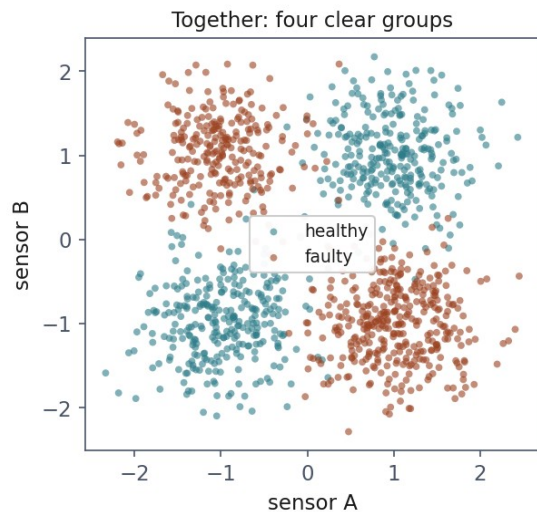
Wrong densities, right boundary



One step away: when the signal is an interaction

- A second pair of sensors on the same fleet
- Faulty when **exactly one** of the two readings is high
- Both high is the designed high-load mode; both low is idle; one without the other is a mismatch between demand and delivery

What Naive Bayes gets to see



0.404: below the baseline, and below chance

Model	Accuracy
Majority baseline	0.523
Gaussian Naive Bayes	0.404
Logistic regression	0.393
SVM, linear kernel	0.606
k-NN, k = 5	0.967
SVM, RBF kernel	0.972

Its probabilities are not probabilities

- Mean confidence when **correct: 0.567**. When **wrong: 0.555**
- It cannot tell the difference between the two situations
- With features that repeat each other it fails the other way: 0.999 reported, with an accuracy nothing like that
- **The ranking may be useful while the probabilities are not**

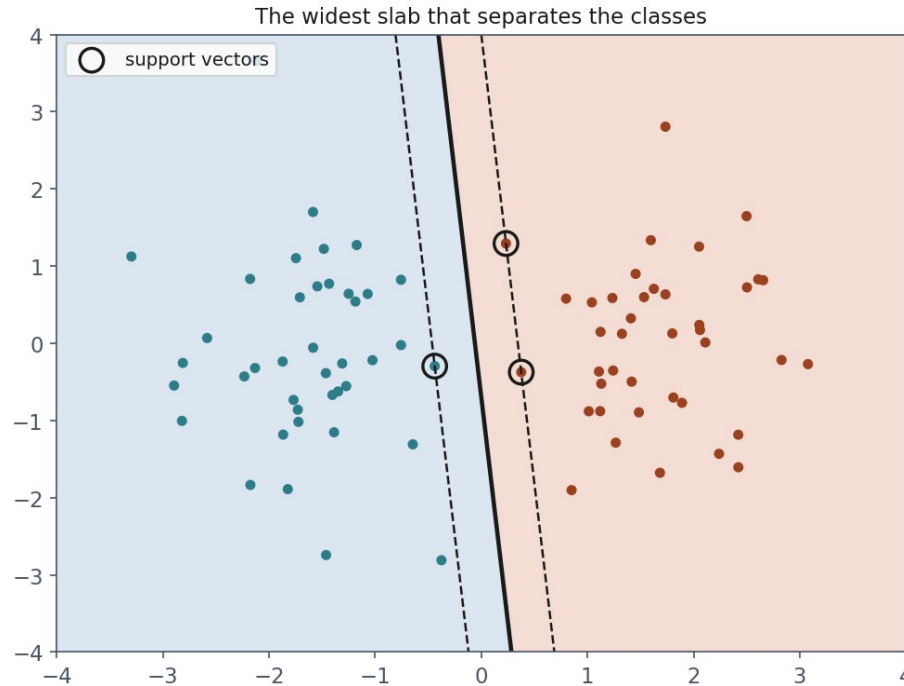
Notebook 2, live

- Where breaking the assumption is cheap, and one step away where it is fatal

The margin: a criterion that is not the error

- On separable data there are **infinitely many** separating lines, and every one has **zero training error**
- Minimising the error therefore cannot choose between them
- The SVM picks the widest **slab**: push it out until it touches the nearest point of each class
- A boundary passing close to a point is one small perturbation from getting it wrong

Three points decide everything



Not separable? Charge for the violations

- Our labels are 4% flipped, so the strict problem has **no solution at all**
- Let each point violate the margin by ξ_i , and charge C for every unit of it
- $\xi_i = 0$ outside the slab, between 0 and 1 inside it, above 1 on the wrong side
- That is the **hinge loss**: exactly zero for a point with room to spare

The hinge stops at zero



A wide slab, minus what the violations cost

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_i \xi_i$$

C is the price of a training error

- **Large C**: violations are expensive, so the model contorts to classify everything. Narrow margin, low bias, high variance
- **Small C**: a wider, calmer boundary, at the cost of some errors
- The same dial as k in k -NN and λ in lesson 3, in a third costume
- Note the direction: **large C means less regularisation**

The best straight line is still a straight line

Model	Accuracy	Support vectors
SVM, linear kernel	0.613 $\pm$ 0.000	947 of 1,200 (79%)
SVM, RBF kernel	0.947 $\pm$ 0.005	278 of 1,200 (23%)

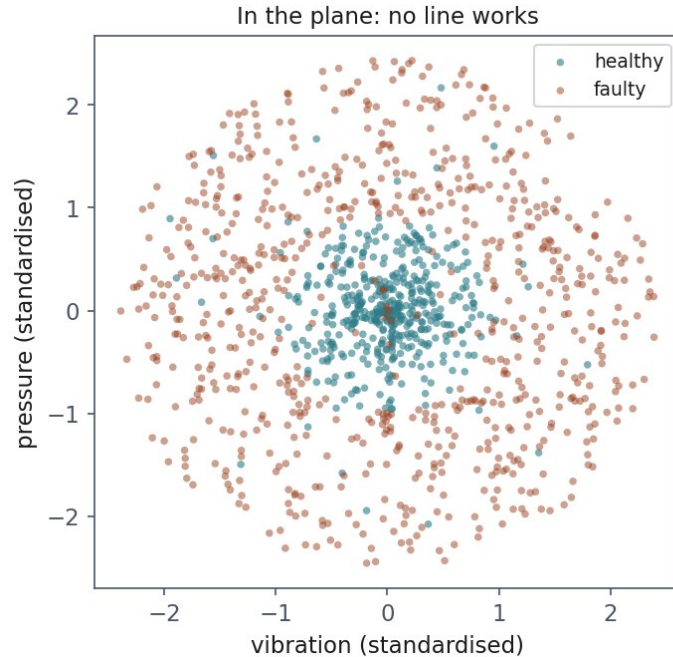
A high support-vector fraction is a free warning

- The linear model needs **79%** of the training set to define its boundary
- The RBF model needs **23%**
- Almost every point sits on or inside the margin: there is no slab that separates anything
- The count falls out of `fit` at no cost, and you should look at it

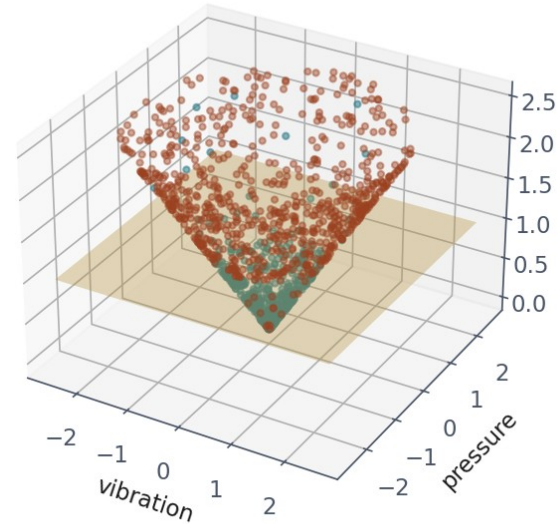
The kernel trick: they needed different coordinates

- Healthy pumps in a disc, faulty ones around it: no line works **in the plane**
- Add a third coordinate: the **distance from the design point**
- Healthy pumps rise a little, faulty ones a lot, and a flat plane separates them perfectly
- The classes were always separable. They were measured in the wrong coordinates

The same pumps, lifted



Lifted: a flat plane does
height = distance from the design point



And the map is never computed

- The solution depends on the data **only through inner products** between pairs of points
- So we need a function returning the inner product **in the new space**, while computing only with the original one
- The radial basis function is scikit-learn's default: one exponential per pair, for an **infinite-dimensional** space

The kernel trick by hand: $25 = 25$

Route	Computation	Result
Lift, then multiply	$(1, 2.83, 4) \cdot (9, 4.24, 1) = 9 + 12 + 4$	25
Never lift	$((1, 2) \cdot (3, 1))^2 = 5^2$	25

The radial basis function, written out

$$K(x, x') = \exp(-\gamma \|x - x'\|^2)$$

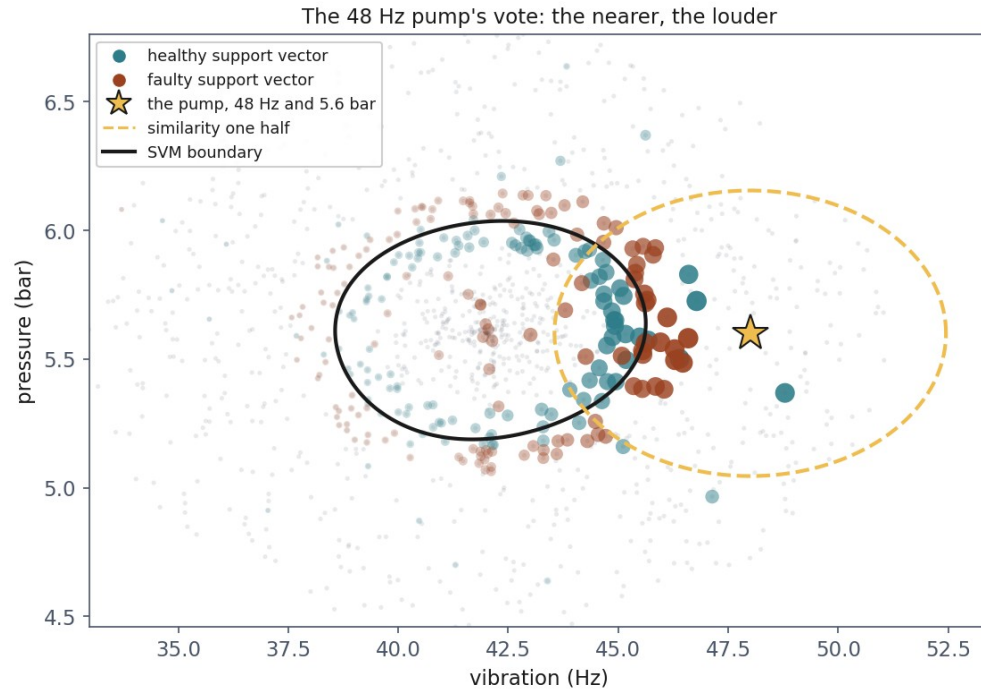
Training: one weight per pump

- Weights adjusted **two at a time**; no learning rate, same answer every run
- **922** pumps: weight 0, forgotten
- **268** at the ceiling $C = 1$, **10** on the edge: the 278 support vectors
- The model: 278 points, their weights, and $b = +1.105$

The SVM on 48 Hz, 5.6 bar: +1.883

Term	Value
votes from faulty support vectors	+37.455
votes from healthy support vectors	-36.677
intercept b	+1.105
total	+1.883

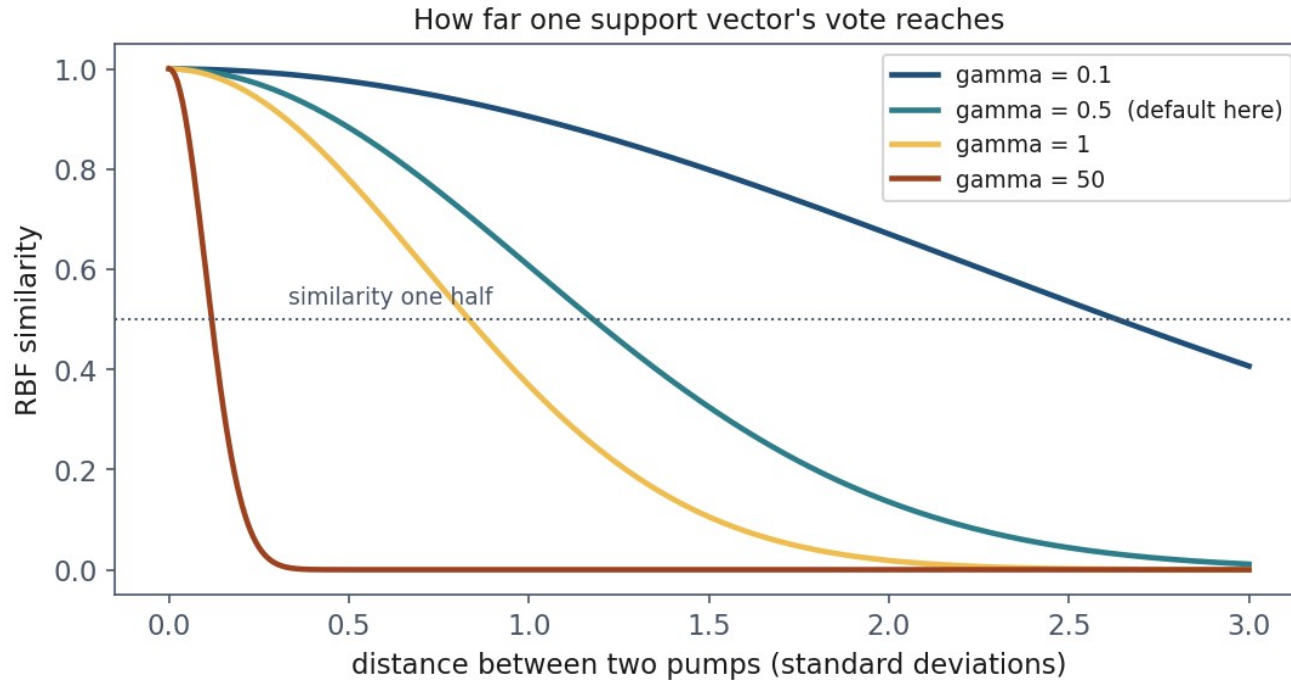
One pump's vote, drawn



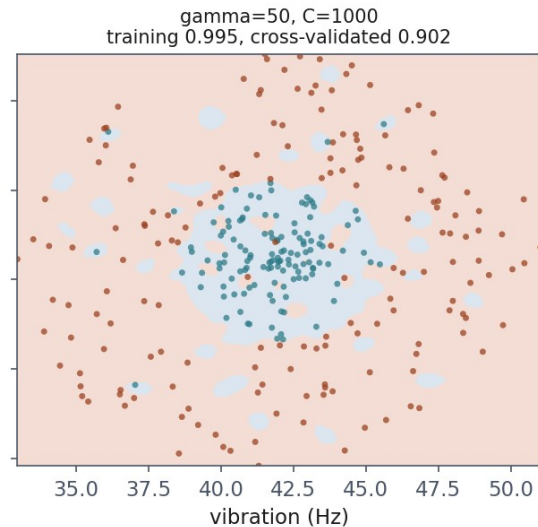
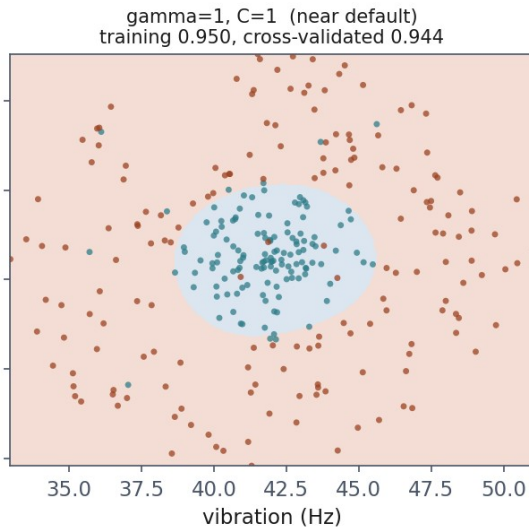
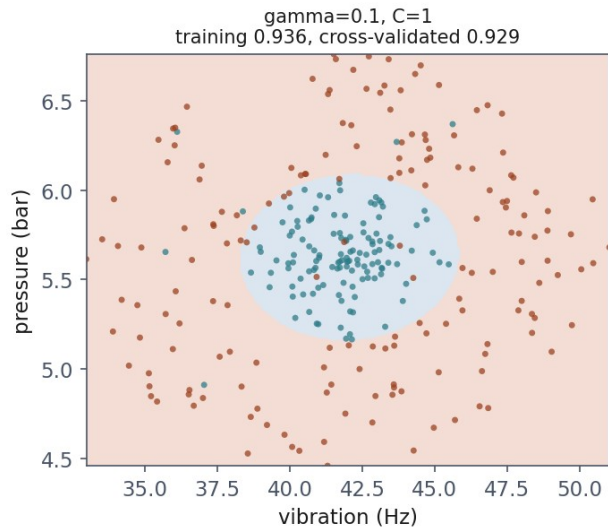
The highest training score, the lowest honest one

Setting	Training	Cross-validated
$\gamma = 0.1, C = 1$	0.936	0.929
$\gamma = 1, C = 1$	0.950	0.944
$\gamma = 50, C = 1000$	0.995	0.902

How far one vote reaches



Overfitting you can see



Notebook 3, live

- Margins, support vectors, and what γ does to the boundary

The three compared, on identical data

Model	Accuracy
Majority baseline	0.613
Logistic regression (lesson 4)	0.613
SVM, linear kernel	0.613
Gaussian Naive Bayes	0.933
k-NN, k = 5	0.944
SVM, RBF kernel	0.947
<i>noise ceiling</i>	≈ 0.96

How to choose, in practice

Situation	Reach for
Few features, plenty of data, odd-shaped boundary	k-NN, or an RBF SVM
Very many features, little data	Naive Bayes
Need a calibrated probability	Not Naive Bayes
Model must be small, or fast at prediction	SVM, not k-NN
The signal is an interaction between features	k-NN or a kernel

What to take away

- **No line separates the pumps:** 0.613, the base rate
- **k is the bias-variance dial**, made visible
- **In 100 dimensions the nearest point is 70% as far as the farthest**
- **Naive Bayes:** 0.933 where independence holds, **0.404** where it does not
- **Choosing the family beats tuning the wrong one:** 0.613 to 0.947

Homework

- **Exercise 6**, discussed at the start of **Friday 13 November**
- A dataset, three families, and a defence of which one you chose